COMPUTER HARDWARE AND SOFTWARE

LECTURE 2

LEARNING OBJECTIVES

1. Understand how data is represented to a computer.
2. Identify several types of input devices and explain their functions.
3. Explain the functions of the primary hardware components found inside the system unit, namely the motherboard, the CPU, and the memory.
4. List several output devices and explain their functions.

LEARNING OBJECTIVES

5. Understand the difference between storage and memory, as well as between a storage device and a storage medium.
6. Name several types of storage systems and explain the circumstances under which they are typically used.
7. Describe the purpose of communications hardware.
8. Understand basic software concepts and commands.

OVERVIEW

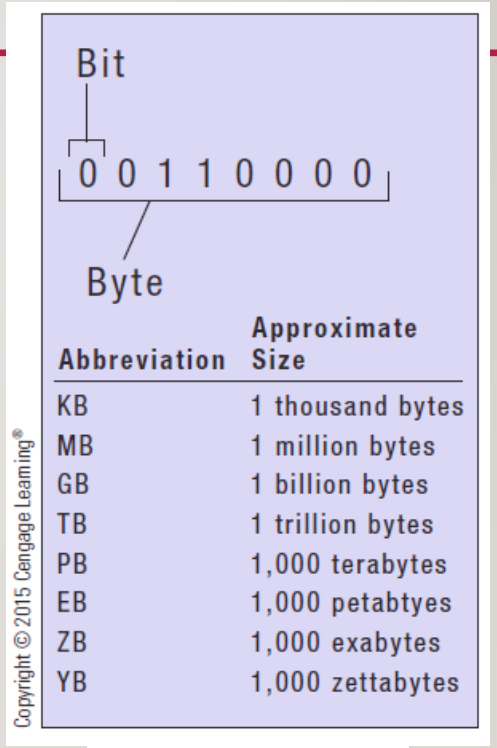
- This Lecture covers:
  - What data is and how it is represented to a computer
  - The hardware that makes up a computer system
  - Various types of hardware used for input, processing, output, storage, and communications
  - Some basic software concepts and operations

5 DIGITAL DATA REPRESENTATION

- Most digital computers are binary computers
  - Understand only two states, off and on
  - Off and on are represented by the numbers 0 and 1
- All data processed by a binary computer must be in binary form
- The computer translates input into the form needed by the computer and translates output into the form needed by the user

6 DIGITAL DATA REPRESENTATION

- Bits and Bytes
  - Bit
    - Smallest unit of data a binary computer can recognize
    - Typically represented by 1s and 0s
    - Derived from the term "binary digits"
    - Generally grouped together to form larger pieces of data
  - Byte
    - Eight bits
    - Prefixes are used to represent larger amounts of data (KB, MB, etc.)



Abbreviation	Approximate Size
KB	1 thousand bytes
MB	1 million bytes
GB	1 billion bytes
TB	1 trillion bytes
PB	1,000 terabytes
EB	1,000 petabtyes
ZB	1,000 exabytes
YB	1,000 zettabytes

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FIGURE 2-2
Bits and bytes. Document size, storage capacity, and memory capacity are all measured in bytes.

7 DIGITAL DATA REPRESENTATION

- Numbering Systems and Coding Systems
  - Numbering system
    - Way of representing numbers
    - Decimal numbering system use ten symbols (0-9) to represent all possible numbers
    - Binary numbering system uses only two symbols (0 and 1) to represent all possible numbers
    - In both systems, the position of each digit determines the power, or exponent, to which the base number (10 for decimal or 2 for binary) is raised

DIGITAL DATA REPRESENTATION

- Coding system
  - Used to represent text-based data
  - ASCII (American Standard Code for Information Interchange)
    - Traditionally used with personal computers
    - 7-digit (7-bit) code
      - Several different 8-bit extended versions of ASCII contain additional symbols not included in the 7-bit ASCII code
      - An 8-bit code can represent 256 characters
    - Limited to only the Latin alphabet used with the English language

9 DIGITAL REPRESENTATION OF DATA

- Unicode
  - Universal standard used to represent text-based data written in any ancient or modern language
  - Uniquely identifies each character using 0s and 1s, regardless of which language, program, or computer is being used
  - Longer code—1 to 4 bytes (8 to 32 bits) per character
  - Can represent over one million characters
  - Can be used worldwide with consistent results

DIGITAL REPRESENTATION OF DATA

CHARACTER	ASCII
0	00110000
1	00110001
2	00110010
3	00110011
4	00110100
5	00110101
A	01000001
B	01000010
C	01000011
D	01000100
E	01000101
F	01000110
+	00101011
!	00100001
#	00100011

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FIGURE 2-3
Some extended ASCII code examples.



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FIGURE 2-4
Unicode. Many characters, such as these, can be represented by Unicode but not by ASCII.

11 QUICK QUIZ

1. Another way to say "one million bytes" is
   a. one kilobyte
   b. one gigabyte
   c. one megabyte
2. True or False: ASCII is the universal standard used to represent text-based data written in any ancient or modern language.
3. The _____ numbering system is used by computers to perform mathematical computations.

Answers:

1) c; 2) False; 3) binary

12 INPUT HARDWARE

- Input Device
  - Any piece of hardware used to perform data input
- Keyboard
  - Used to type data into a computer
  - Can be integrated, wired, or wireless
  - Nearly always used with desktop and notebook computers
  - Many mobile phones have an integrated keyboard



FIGURE 2-5
Slide-out keyboards.

13 INPUT HARDWARE

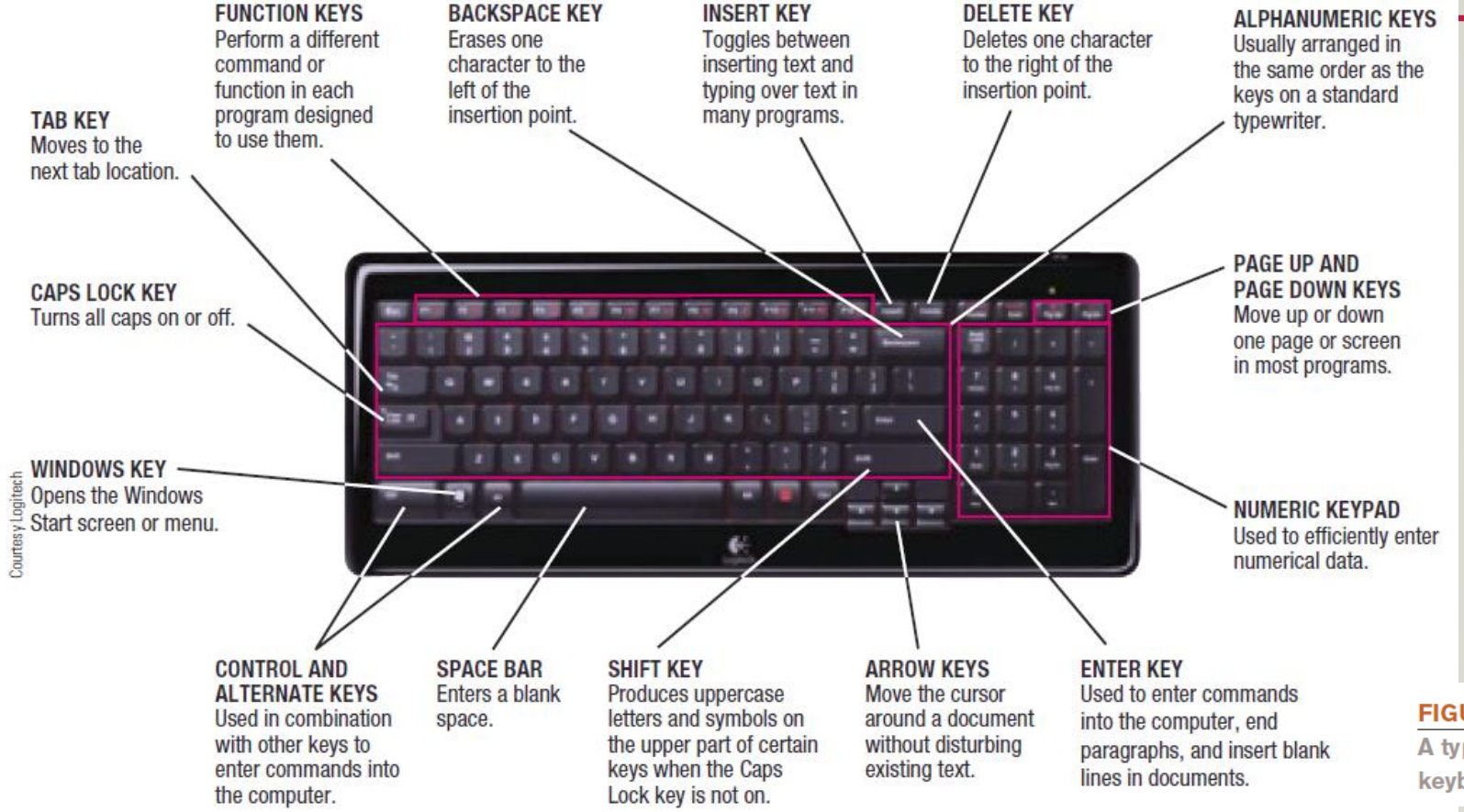


FIGURE 2-6
A typical desktop keyboard.

14 INPUT HARDWARE

- Pointing Devices
  - Used to select objects, issue commands, etc. by pointing to and selecting objects
  - Two most common are the mouse and the pen stylus
  - Mice
    - Nearly always used with desktop computers; can be used with notebook computers
      - Most are either optical or laser mice
      - Used to start programs; open, move around, and edit documents; draw or edit images

15 INPUT HARDWARE

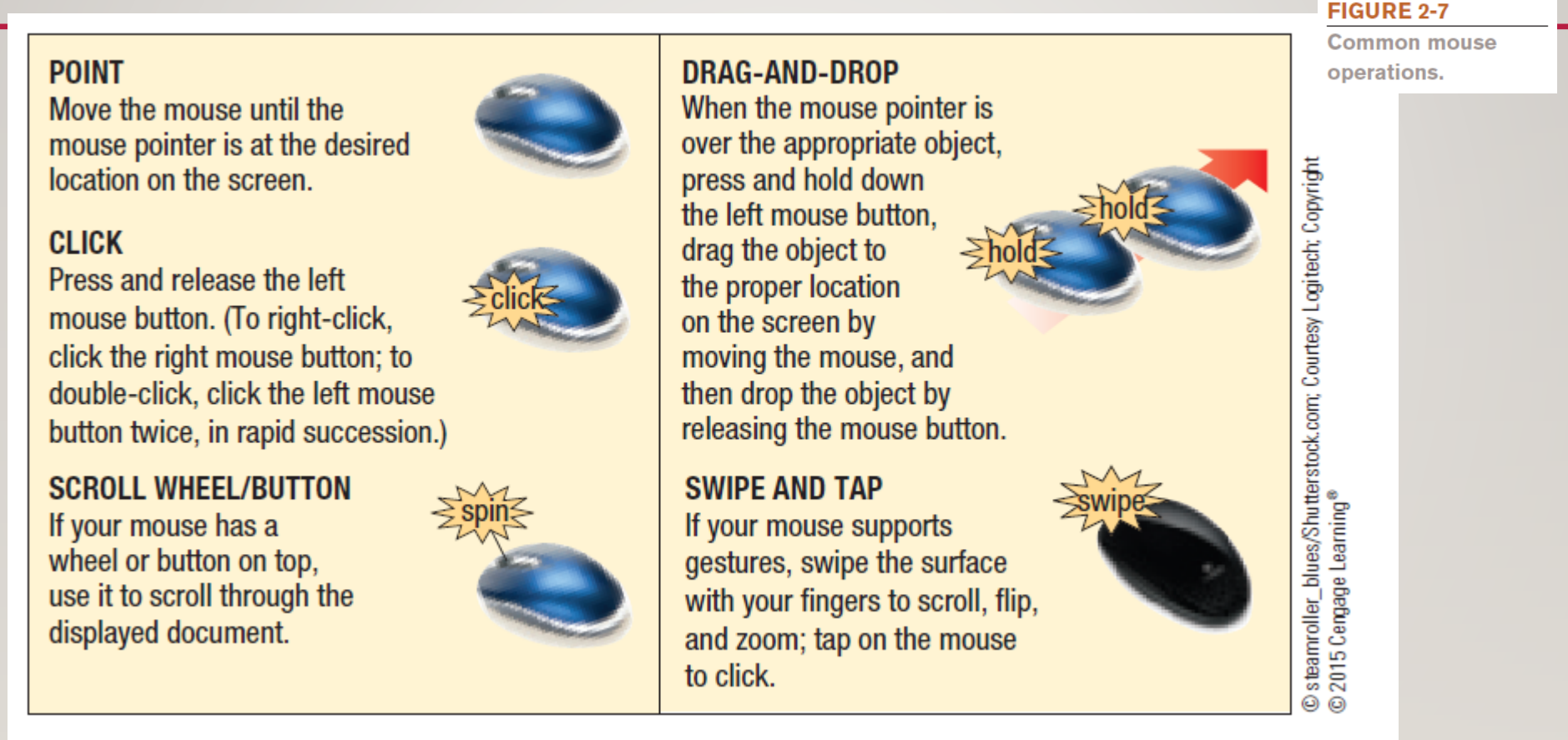


FIGURE 2-7
Common mouse operations.

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16 INPUT HARDWARE

- Pens/Styluses
  - Used to select items and/or write electronically on the screen
  - Can be used with personal computers, mobile devices, and tablet computers
  - Increasing used for photography, graphic design, animation, and document processing

FIGURE 2-8
Examples of digital pen use.



17 INPUT HARDWARE

- Touch Screens
  - Allows the user to touch the screen with his or her finger to provide input
  - Also used in consumer kiosks and POS systems
  - Many are multi-touch
  - Convenient for most users
  - Not accessible for blind individuals and those with limited mobility
  - Other pointing devices
    - Control buttons and wheels, trackballs, touch pads, and gaming devices

18 INPUT HARDWARE



FIGURE 2-9
Examples of other common pointing devices.

19 INPUT DEVICES

- Scanners, Readers, and Digital Cameras
  - Scanners and Readers
    - Used to capture data from a source document and input it into the computer
    - Commonly called optical scanners
    - Documents input as a single graphical image
      - Editing text within the image requires the use of OCR software in conjunction with the scanner
    - Flatbed scanner is the most common
    - Portable scanners used to capture text and other data while on the go

20 INPUT DEVICES



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FLATBED SCANNERS
Used to input photos, sketches, slides, book pages, and other relatively flat documents into the computer.



Courtesy United Parcel Service of America, Inc.

PORTABLE BARCODE READERS
Used to read barcodes when portability is needed.



Courtesy of teamaxess.com



Courtesy MasterCard Worldwide

STATIONARY RFID READERS
Used to read RFID tags, such as to automatically open ski lift entry gates for valid lift ticket holders at a ski resort in Utah (left) and to process payments via RFID-enabled credit cards (right).

FIGURE 2-10
Scanners and readers transform data from physical form to digital form.

21 INPUT DEVICES

- Resolution
  - Quality of scanned images
  - Usually measured in dpi—number of dots per inch
- Barcode Readers
  - Read barcodes, which are optical codes that represent data
  - Most common barcodes are UPC labels and ISBN numbers
  - Businesses and organizations can create custom barcodes for unique needs
    - FedEx, UPS, USPS
    - Hospitals, researchers, law enforcement



22 TECHNOLOGY AND YOU BOX

Mobile Ticketing

- Download ticket to mobile device to use for admittance
- Tickets for movies, sporting events, concerts, etc.
- Boarding passes
- Custom tickets



23 INPUT DEVICES

- RFID (radio frequency identification) Readers
  - RFID technology stores, reads, and transmits data found in RFID tags
  - RFID tags can be attached to many different types of objects and are read by radio waves
  - Data in the tags can be unique so that each item containing an RFID tag can be individually identified
  - Growth in retail industry has been slower than expected due to cost constraints and privacy/security issues

24 INPUT DEVICES

- Biometric Readers
  - Read biometric data (measurable biological characteristics)
    - Fingerprints, hand geometry, face, iris, voice
  - Can be stand alone or built into another piece of hardware
  - Other types of readers
    - Optical mark readers
    - Optical character readers
    - Magnetic ink character recognition readers



FIGURE 2-12
Biometric readers.

25 INPUT DEVICES

- Digital Cameras
  - Record images on digital storage media such as flash memory cards, hard drives, DVD discs, etc.
  - Digital still cameras
  - Digital video cameras
    - Digital camcorders
    - PC video cameras
      - Used for webcams, videoconferencing, video phone calls, etc.

FIGURE 2-13
Digital cameras.



Courtesy Sony Electronics Inc.; ©MasasiS/Shutterstock.com; Courtesy Kingston Technology Company, Inc.

Virtually all digital cameras let you display and erase images.

Most cameras use removable storage media in addition to, or instead of, built-in storage.

CONSUMER DIGITAL STILL CAMERAS
Typically store photos on flash memory media.



Courtesy Sony Electronics Inc.

CONSUMER DIGITAL CAMCORDERS
Typically store video on a built-in hard drive (as in this camera) or on DVD discs.

26 INPUT DEVICES

- Other Input Devices
  - Microphones and headsets
    - Used for voice input
      - Commands, dictation, podcasts, etc.
  - MIDI keyboards, keyboard and guitar controllers
    - Used to input original music
  - Adaptive input devices
    - Designed for users with physical disabilities
  - Ergonomic hardware
    - Designed to lessen the physical effects of computer use

27 QUICK QUIZ

1. An optical mouse is _____.
    a. the same as a wireless mouse
    b. a mouse that tracks movements with light instead of a ball
    c. a mouse that contains a scroll wheel on the top
2. True or False: Flatbed scanners are used to read biometric data, such as fingerprints, hand geometry, veins, and irises.
3. _____ input devices are designed for users with disabilities.

Answers:

1) b; 2) False; 3) Adaptive

PROCESSING HARDWARE AND OTHER HARDWARE INSIDE THE SYSTEM UNIT

- System Unit
  - Main case of the computer
  - Components
    - Processing hardware, storage devices
    - Power supply, cooling fan
    - Interfaces used to connect peripheral devices
- The Motherboard
  - The main circuit board inside the system unit
  - All devices used with a computer connect in one way or another to the motherboard (directly, via port, etc.)

PROCESSING HARDWARE AND OTHER HARDWARE INSIDE THE SYSTEM UNIT

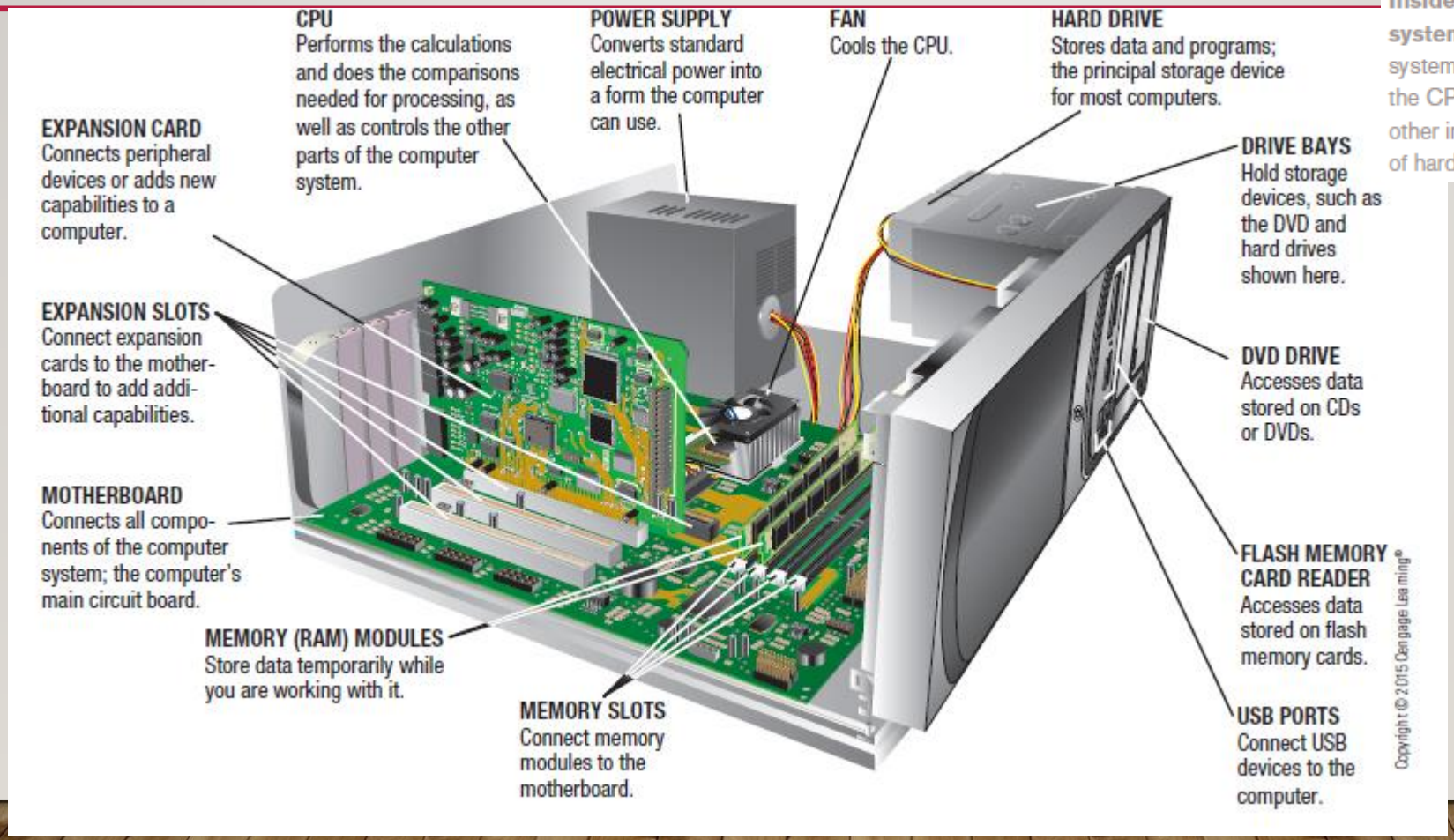


FIGURE 2-14
Inside a typical system unit. The system unit houses the CPU, memory, and other important pieces of hardware.

PROCESSING HARDWARE AND OTHER HARDWARE INSIDE THE SYSTEM UNIT

- Ports
  - Connectors exposed through the exterior of the system unit case
  - Either built into the motherboard or created via expansion card
  - Used to connect external devices to the computer (monitors, keyboards, mice, printers, etc.)
  - Portable computers have fewer ports

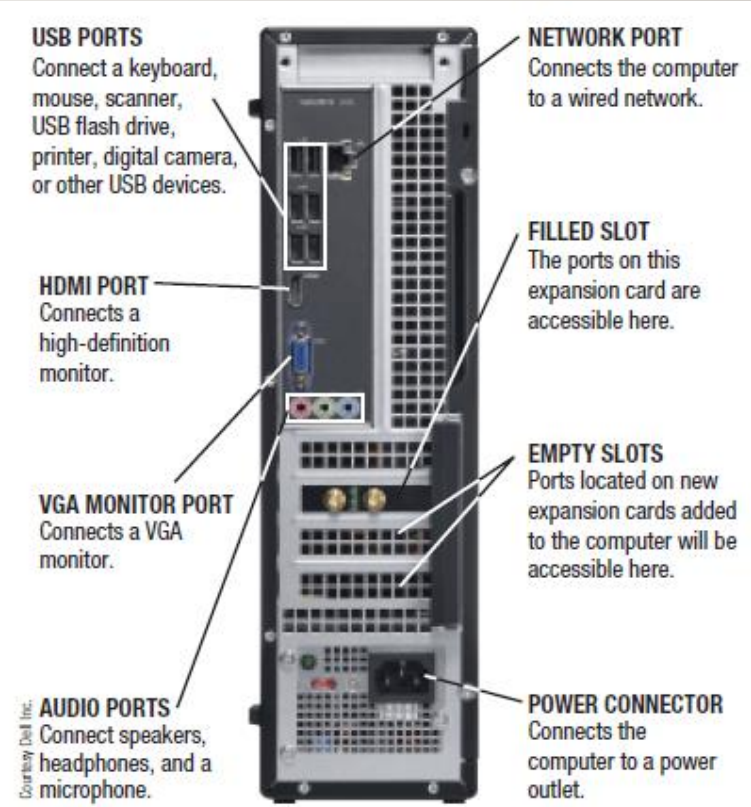


FIGURE 2-15
Ports are used to connect external devices to the motherboard.

PROCESSING HARDWARE AND OTHER HARDWARE INSIDE THE SYSTEM UNIT

- Processors
  - Consist of a variety of circuitry and components that are packaged together and connected directly to the motherboard
- The Central Processing Unit (CPU); also called microprocessor
  - Arithmetic/logic unit
    - Performs arithmetic involving integers and logical operations
  - Control unit
    - Coordinates and controls the actions taking place within the CPU

PROCESSING HARDWARE AND OTHER HARDWARE INSIDE THE SYSTEM UNIT

- Most personal computers today use CPUs manufactured by Intel or Advanced Micro Devices (AMD) and are multi-core
  - Dual-core CPU
    - Contains the processing components (cores) of two independent processors on a single CPU
  - Quad-core CPU
    - Contains four cores

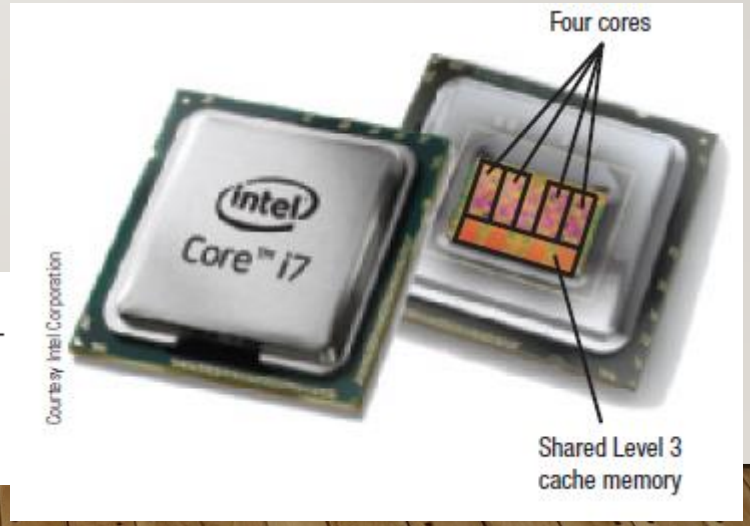


FIGURE 2-16
CPUs. CPUs today typically have multiple cores.

PROCESSING HARDWARE AND OTHER HARDWARE INSIDE THE SYSTEM UNIT

- Measurements of CPU processing speed
  - CPU clock speed is measured in megahertz (MHz) or gigahertz (GHz)
  - Higher clock speed means more instructions can be processed per second than the same CPU with a lower clock speed
  - Other factors can affect the overall processing speed of a computer
    - Number of cores
    - Amount of memory
    - Speed of external storage devices

34 INSIDE THE SYSTEM UNIT

- The Graphics Processing Unit (GPU)
  - Takes care of the processing needed to display images (including still images, animations) on the screen
  - Can be located on the motherboard, on a video graphics board, on in the CPU package

INSIDE THE INDUSTRY BOX

GPUs and *Transformers: The Ride 3D* at Universal Studios

- Uses a 2,000 foot-long track, 14 huge screens, and 34 projectors
- Motion is synchronized with the action
- Images are 3D, 4K
- Used GPUs to see 3D animations in real time as they were being developed



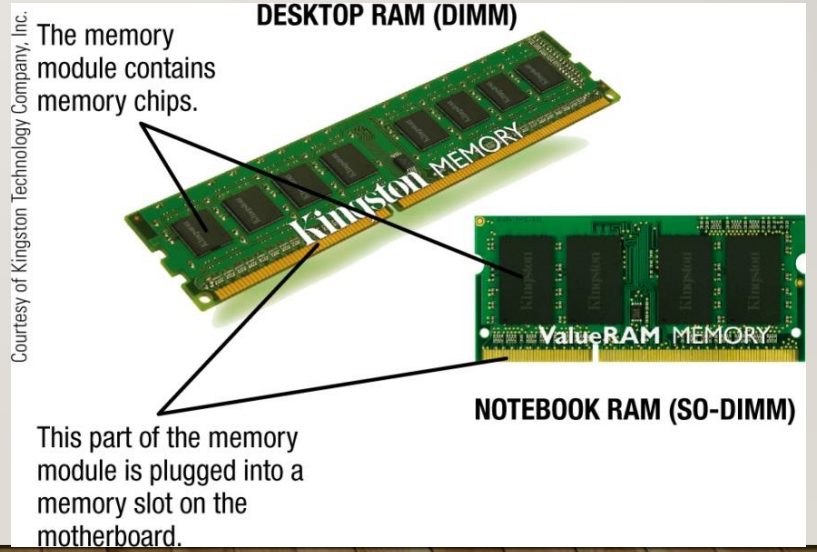
PROCESSING HARDWARE AND OTHER HARDWARE INSIDE THE SYSTEM UNIT

- Memory
  - Refers to chip-based storage used by the computer – usually the amount of the computer's main memory (random access memory or RAM)
  - RAM
    - Temporary (volatile) storage used by the computer
    - Used to store essential parts of the operating system while computer is running
    - Stores the programs and data currently being used by the computer
    - Content is lost when computer is shut down
    - Data is deleted from RAM when it is no longer needed

PROCESSING HARDWARE AND OTHER HARDWARE INSIDE THE SYSTEM UNIT

- Consists of circuits etched onto chips arranged onto circuit boards called memory modules
- Capacity is measured in bytes
  - Computers must have enough RAM to run the necessary applications, as well as work efficiently
  - More RAM allows more programs to run at one time
  - Most PCs today come with at least 512 MB of RAM

FIGURE 2-17
RAM memory modules.



PROCESSING HARDWARE AND OTHER HARDWARE INSIDE THE SYSTEM UNIT

- Other types of computer memory
  - Cache memory
    - Used to speed up the processing speed by storing the data that may be needed next by the CPU in handy locations
    - Usually built into the CPU
    - Level indicates the order the cache memory is accessed (L1 = fastest and checked first, L2, L3)
  - Registers
    - Used by the CPU to temporarily store data and intermediary results during processing
    - Fastest type of memory

PROCESSING HARDWARE AND OTHER HARDWARE INSIDE THE SYSTEM UNIT

- ROM
  - Read-only memory
  - Nonvolatile chips that permanently store data or programs accessed by the computer
  - Attached to the motherboard like RAM
  - Data and programs are retrieved by the computer when needed
  - Cannot be written over
  - Data is not erased when the computer is shut down

PROCESSING HARDWARE AND OTHER HARDWARE INSIDE THE SYSTEM UNIT

- Flash memory
  - Nonvolatile chips into which data can be stored and retrieved
  - Flash memory chips have begun to replace ROM for storing system information, as it can be updated as needed
  - Also used in flash memory storage systems

41 QUICK QUIZ

1. Which type of memory is erased when the power goes out?
    a. ROM
    b. RAM
    c. flash memory
2. True or False: The CPU can also be called the motherboard.
3. A CPU with four cores is called a(n) _____ CPU.

Answers:

1) b; 2) False; 3) quad-core

42 OUTPUT HARDWARE

- Display Devices
  - Most common type of output device
  - Displays output on some type of screen
  - Monitor
    - Display device for a desktop computer
  - Display screen
    - Built-in display device
    - Used with notebook computers, mobile phones, etc.
    - Used with many consumer products

FIGURE 2-18
Many consumer products today have a display screen.



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MEDIA TABLETS



Courtesy Sony Electronics Inc.

SMART WATCHES

43 OUTPUT HARDWARE

- CRT monitor
  - Traditional monitor for desktop computers
  - Uses cathode-ray tube technology
  - Very large and bulky
  - Less common today
- Flat-panel displays
  - Form images by manipulating electronically charged chemicals or gasses sandwiched between panes of glass or other transparent surfaces
  - Take up less space and consume less power than CRT monitors

44 OUTPUT HARDWARE

- Thinner and lighter than CRTs
- Used with most computers and in mobile phones and other mobile devices

- Pixel
  - Smallest colorable area in an electronic image
  - Number of pixels used on a display screen determines the screen resolution
  - With high resolution, more information can fit on the screen, but everything will be displayed smaller than with a lower resolution

45 OUTPUT HARDWARE

- Multiple monitors
  - Can be used with both desktop and portable computers
  - Typically extends the desktop but can duplicate it



FIGURE 2-19
Multiple monitors.
Can be used to extend a desktop, which can increase productivity.

46 OUTPUT DEVICES

- Printers
  - Produce hard copy – a permanent copy of the output on paper
  - Personal printers designed to be connected to a single computer
  - Network printers designed to be shared by multiple users via a network
  - Impact printers
    - Strike paper to transfer ink to paper

47 OUTPUT DEVICES

- Non-impact printers
  - Form images without touching the paper
  - Most common type of printer today
  - Much quieter than impact printers
  - Produce higher quality images
- Can be color or black-and-white
- Printer quality is measured in dots per inch (dpi)
- Printer speed is measured in pages per minute (ppm)

FIGURE 2-20
Color printing.
Requres multiple cartridges or cartridges that contain multiple colors.



© Paul Broadbent/Shutterstock.com

48 OUTPUT DEVICES

- Laser Printers
  - Standard for business documents
  - Come in personal and network versions
  - Can be color or black-and-white
  - Faster and of better quality than ink-jet printers
  - Use technology similar to photocopy machines (toner powder, lasers, heat, etc.)
  - Resolutions are between 600 and 2,400 dpi
  - Common speeds for personal laser printers range from 10 to 70 ppm

49 OUTPUT DEVICES

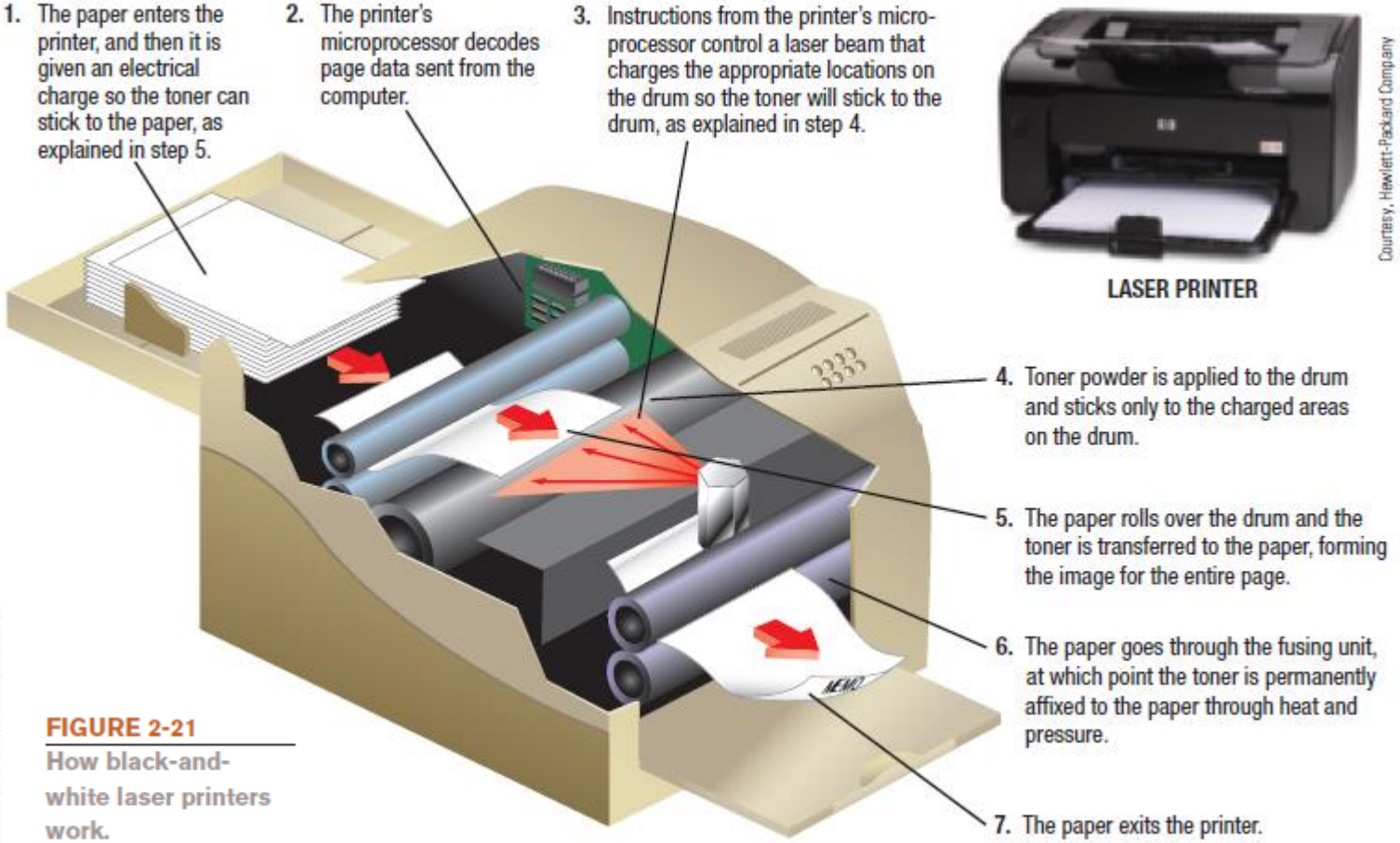


FIGURE 2-21
How black-and-white laser printers work.

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50 OUTPUT DEVICES

- Ink-Jet Printers
  - Spray tiny drops of ink onto the page, one printed line at a time
  - Some print using different-sized ink droplets, multiple nozzles, and varying electrical charges for precision
  - Fairly inexpensive, although replacement ink cartridges can add up
  - Good quality and usually can print in color
  - Printer of choice for home use
  - With special paper, ink-jet printers can print photos

51 OUTPUT DEVICES



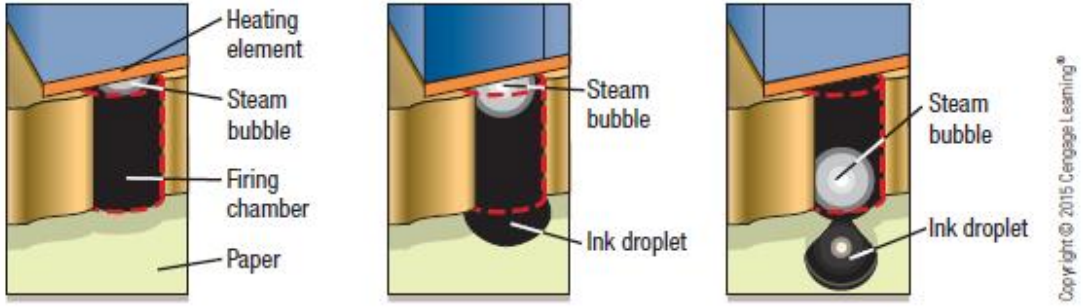
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Each ink cartridge is made up of multiple tiny ink-filled firing chambers; to print images, the appropriate color ink is ejected through the appropriate firing chamber.



Courtesy, Hewlett-Packard Company

INK-JET PRINTER



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1. A heating element makes the ink boil, which causes a steam bubble to form.
2. As the steam bubble expands, it pushes ink through the firing chamber.
3. The ink droplet is ejected onto the paper and the steam bubble collapses, pulling more ink into the firing chamber.

FIGURE 2-22
How ink-jet printers work.

52 OUTPUT DEVICES

- Special-Purpose Printers
  - Photo printers
    - Designed to print photographs
  - Barcode printers
    - Print barcodes
  - Portable printers
    - Designed to be carried with you to print while on the go
  - Plotters and wide-format ink-jet printers
    - Print large documents, such as blueprints and signs
  - 3D printers
    - Form a 3D replica of an object, typically using plastic powder

OUTPUT DEVICES

- Other Output Devices
  - Computer speakers
  - Headphones
  - Headsets
  - Earphones
  - Earbuds

54 QUICK QUIZ

1. Which of the following could possibly use CRT technology?
    a. Smart watch
    b. Desktop monitor
    c. Media tablet
2. True or False: Laser printers can only print in black and white.
3. _____ printers form images with drops of liquid ink.

Answers:

1) b; 2) False; 3) Ink-jet printers

55 STORAGE HARDWARE

- Storage System
  - Used to save documents for future use
  - Non-volatile
- Storage System Characteristics
  - Storage Media and Storage Devices
    - Storage Medium
      - The hardware where the data is actually stored
      - DVD, flash memory card, etc.
    - Storage Devices
      - The device that reads data from or writes data to the storage medium

56 STORAGE HARDWARE

- Internal
  - Located inside the system unit
- External
  - Plugged into an external port on the system unit
- Remote
  - Located on another computer, such as a network server

57 STORAGE HARDWARE

- Storage medium and storage device are often two separate pieces of hardware
- Some systems are permanently sealed together
  - Hard drive
  - USB flash drives
- Are identified by letters of the alphabet or nam… assigned by the computer

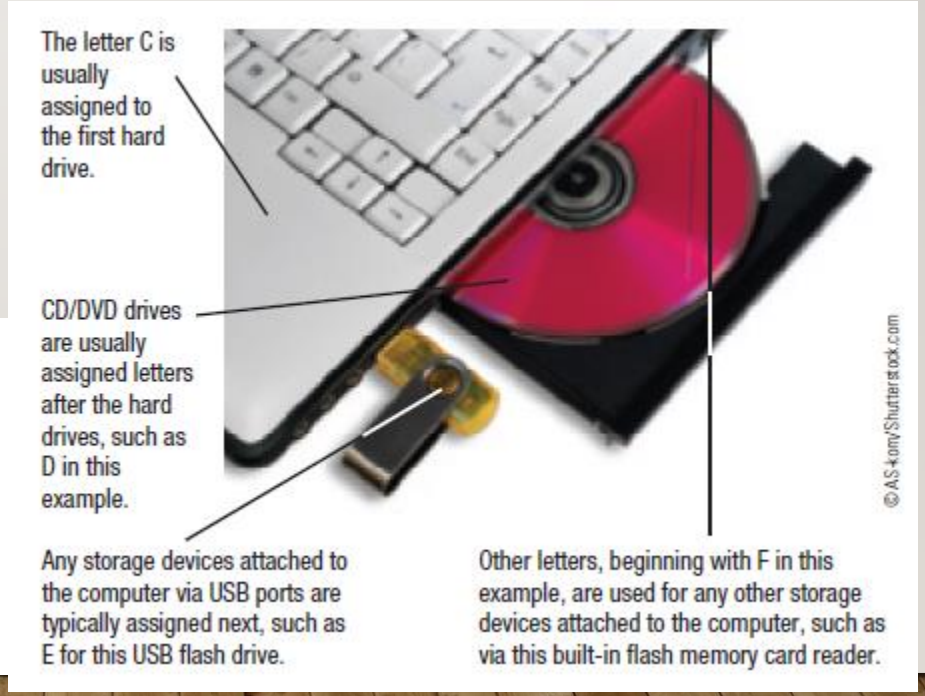


FIGURE 2-23
Storage device identifiers. To keep track of storage devices in an unambiguous way, the computer system assigns letters of the alphabet or names to each of them.

58 STORAGE HARDWARE

- Types of Storage Technology Used
  - Data is usually stored magnetically or optically
    - Magnetic storage systems (hard drives) represent data using different magnetic alignments
    - Optical media (CDs and DVDs) use laser beams that burns permanent marks into the surface of the medium
  - Flash memory storage systems represent data using electrons

STORAGE HARDWARE

- Hard Drives
  - Used to store most programs and data
  - Internal hard drives
    - Not designed to be removed
  - External hard drives
    - Connect to a computer using a USB or FireWire port
  - Both are increasingly coming with built-in encryption



Courtesy Apricorn

FIGURE 2-24
Encrypted hard drives. The data stored on this external hard drive is accessed via a fingerprint scanner.

60 STORAGE HARDWARE

- Magnetic Hard Drives
  - Use metal hard disks organized into tracks and sectors
  - Use read/write heads to store (write) and retrieve (read) data

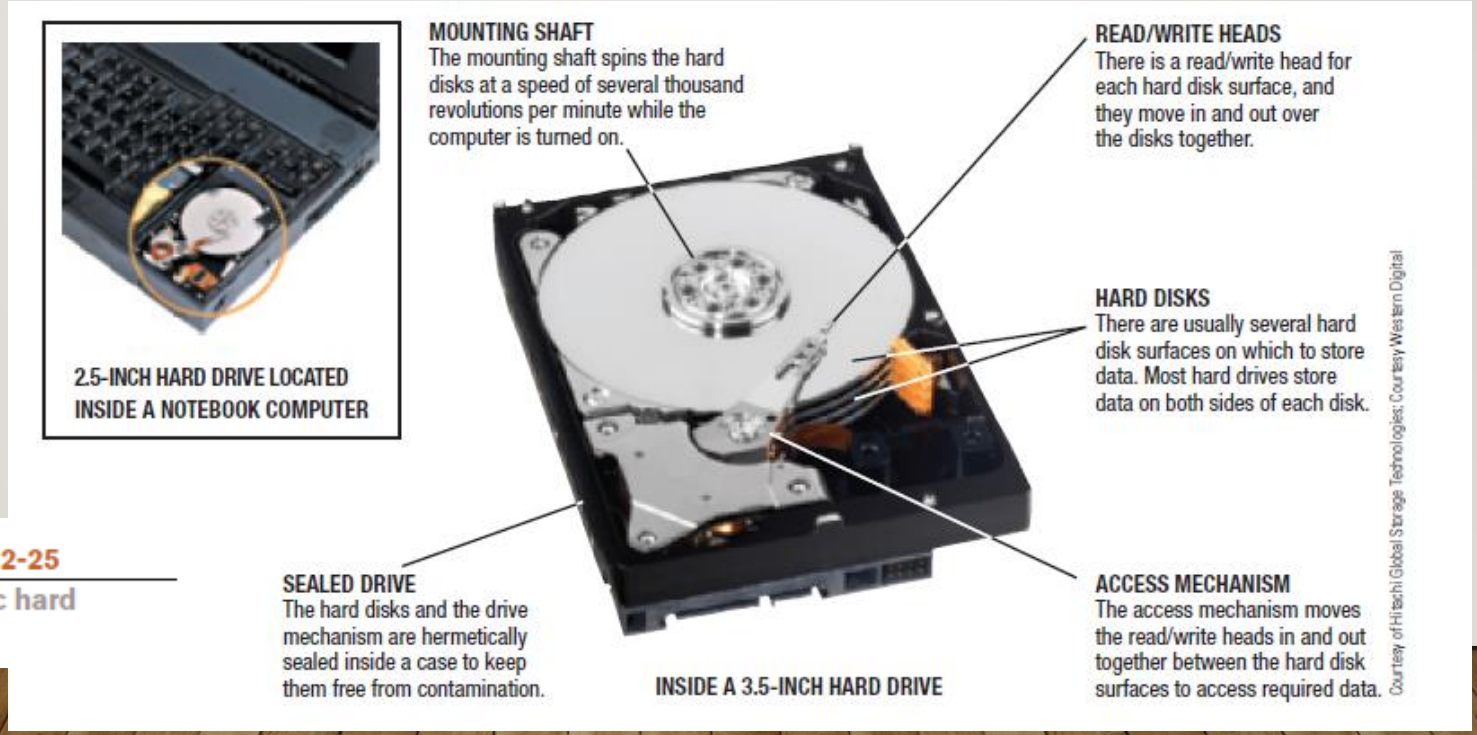


FIGURE 2-25
Magnetic hard drives.

61 STORAGE HARDWARE

- Solid-State Drives (SSDs)
  - Use flash memory technology rather than spinning platters and magnetic technology
  - Use less power and have no moving parts
  - Not subject to mechanical failure and more resistant to shock and vibrations
  - Generate less heat, make no noise, and operate faster
  - Attractive option for portable computers and mobile devices



FIGURE 2-26
Solid-state drives (SSDs).

62 STORAGE HARDWARE

- Hybrid Hard Drives
  - Use a combination of flash memory and a magnetic hard drive
  - Less expensive than an SSD
  - Can extend the battery life of portable computers and mobile devices
  - Can allow encryption and other security measures to be built into the drive

STORAGE HARDWARE

- Internal vs. External Hard Drives
  - Internal Hard Drives
    - Permanently located in the system unit
    - Almost all computers have at least one used to store programs and data
  - External Hard Drives
    - Used to move data from one computer to another
    - Can hold large amounts of data—1 TB to 4 TB
    - Used by individuals to hold digital pictures, digital music, home movies, recorded television shows
    - Portable hard drives are smaller external hard drives

64 STORAGE HARDWARE

- Most external and portable hard drives connect using a USB connection
- Some can connect using a wired or wireless networking connection



Courtesy Western Digital

FULL-SIZED EXTERNAL HARD DRIVES
Are about the size of a 5 by 7-inch picture frame, but thicker; this drive contains two magnetic hard drives, which hold 6 TB total.



Courtesy Transcend Information USA

PORTABLE HARD DRIVES
Are about the size of a credit card, but thicker; this SSD drive holds 256 GB.

FIGURE 2-27
External hard drives.

65 STORAGE HARDWARE

- Optical Discs and Drives
  - Data on optical discs (DVDs, CDs, Blu-ray) is stored and read optically, using laser beams
  - Data can be stored on one or both sides of the disc
  - Discs are divided into tracks and sectors, but use a single grooved spiral track beginning at the center of the disc
  - Lasers create and read marks created on the disc surface called pits and lands
    - With read-only discs, the pits are permanent
    - With recordable or rewritable discs, the pits are represented by changing the disc's reflectivity

STORAGE HARDWARE

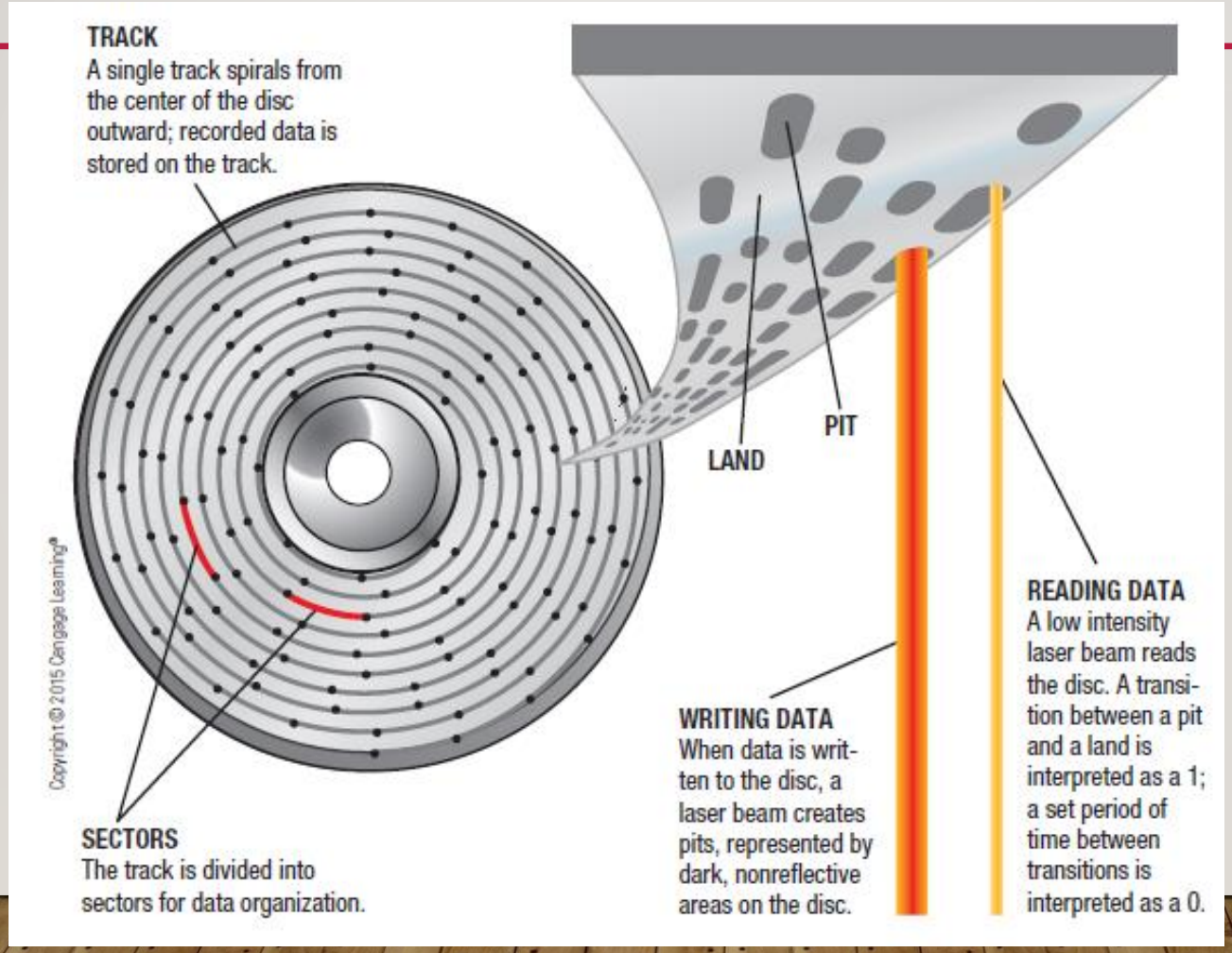


FIGURE 2-28
How recorded optical discs work.

67 STORAGE HARDWARE

- Conventional CDs
  - Use infrared lasers
- Conventional DVDs
  - Use red lasers
- High definition Blu-ray Discs (BDs)
  - Use blue-violet lasers
  - More information can be stored (HD movies, etc.)
  - Data is recorded more compactly
- Discs come in various sizes
- Capacity depends on format, as well as the number of sides and layers, used

68 STORAGE HARDWARE



FIGURE 2-29
Optical discs are available in a variety of sizes, appearances, and capacities.

69 STORAGE HARDWARE

- Read-Only Discs: CD-ROM, DVD-ROM, and BD-ROM Discs
  - Come pre-recorded with commercial products
  - Data cannot be erased, changed, or added to
- Recordable Discs: CD-R, DVD-R, and BD-R Discs
  - Can be written to, but the discs cannot be erased and reused
  - Commonly used for:
    - Backing up files
    - Sending large files to others
    - Creating custom music CDs (CD-R)
    - Storing home movies, digital photos, multimedia files, etc. (DVD-R/+R)
    - Storing high-definition multimedia files (BD-R)

70 STORAGE HARDWARE

- Rewritable Discs: CD-RW, DVD-RW, DVD+RW, and BD-RE Discs
  - Can be written to, erased, and overwritten like magnetic disks
  - The correct drive must be used to write each type of disc
  - Used for similar purposes as recordable discs but are especially appropriate when data can later be erased and the disc reused
- Optical Drive
  - Most computers have one; external drive can be used

STORAGE HARDWARE

- Flash Memory Storage Systems
  - Chip-based storage medium that represents data using electrons
  - No moving parts, so are not as subject to mechanical failure as optical and magnetic media
  - Are more resistant to shock and vibration
  - Consume less power
  - Very small and quiet
  - Rewritable
  - Longer expected life that magnetic media
  - More expensive per MB, but very convenient and widely used with computers, digital cameras, etc.

STORAGE HARDWARE

- Embedded Flash Memory
  - Chips are embedded in products to provide built-in storage
    - Portable digital media players
    - Digital cameras
    - Media tablets
    - Mobile phones
    - Sunglasses, wristwatches, etc.

FIGURE 2-31
Embedded flash memory.



73 STORAGE HARDWARE

- Flash Memory Cards and Readers
  - Small card containing flash memory chips, a controller chip, other electrical components, and metal contacts to connect the card to the device or reader being used
  - Read by flash memory card readers (built-in or standalone)
  - Commonly used with:
    - Desktop and portable computers
    - Digital cameras, portable digital media players, mobile phones, and other mobile devices
  - Variety of formats (SD cards, SDHC cards, CompactFlash cards, Memory Stick, etc.) and sizes

74 STORAGE HARDWARE

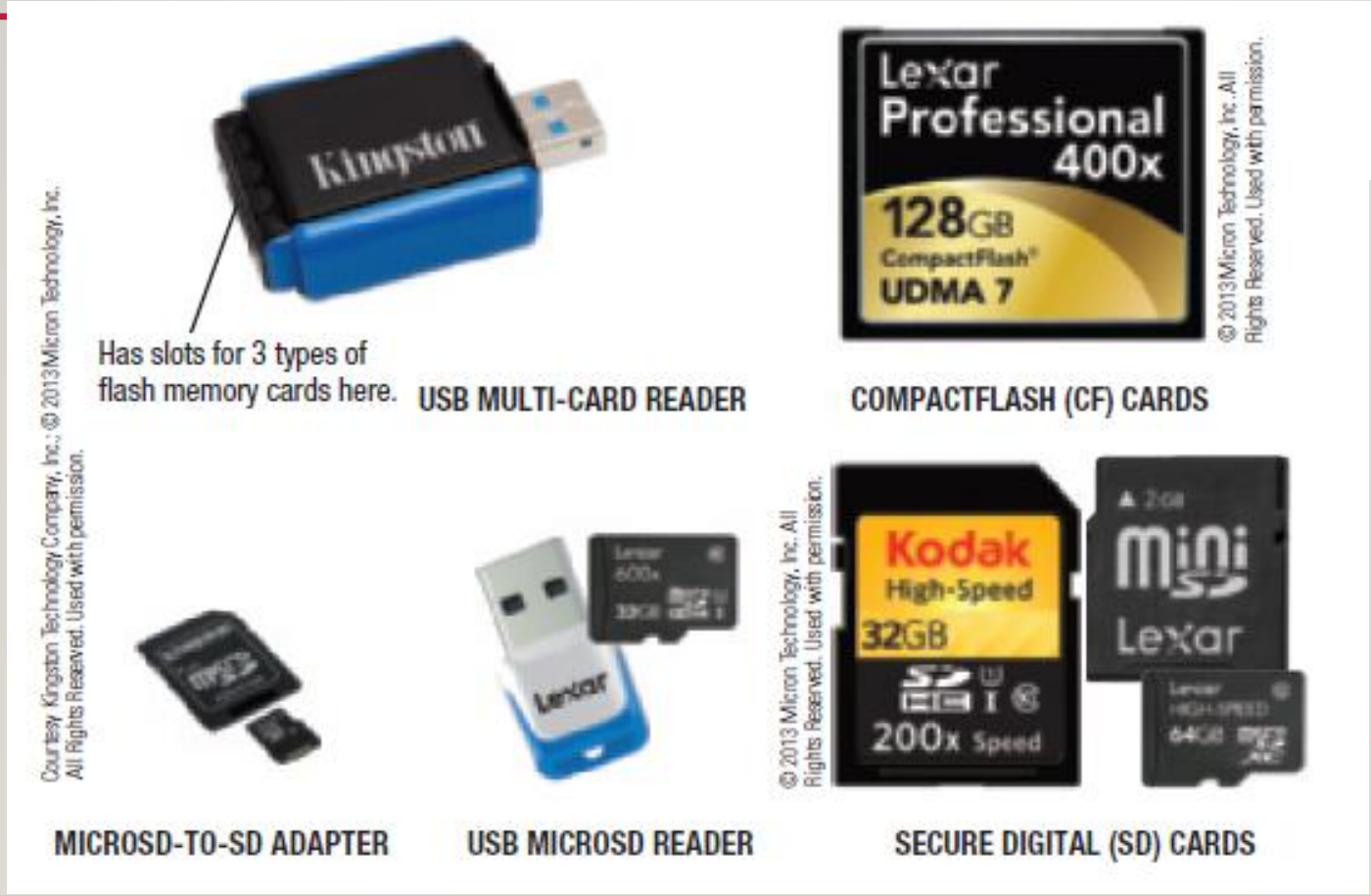


FIGURE 2-32
Some flash memory cards, readers, and adapters.

STORAGE HARDWARE

- USB Flash Drives
  - Flash memory media in self-contained unit that uses a USB interface
  - Also called USB flash memory drives, thumb drives, and jump drives
  - Are powered via the USB port
  - Very portable; can be carried in a pocket or on a keychain
  - Wide range of colors, sizes, appearances
  - Capacity ranges from 4 GB to 1 TB
  - Widely used to transfer files from one location to another

STORAGE HARDWARE

- Can also be used to lock a computer or issue passwords
- Can include biometric features



Courtesy Kingston Technology Company, Inc.

FIGURE 2-33
USB flash drives. Are often used to store data and transfer files from one device to another.

77 HOW IT WORKS BOX

More Storage for Your Tablet

- Media tablets often have 64 GB of storage
- To extend storage, can transfer content to and from PCs
- Easier is to use a wireless hard drive
- Need the appropriate app



78 STORAGE HARDWARE

- Other Types of Storage Systems
  - Network and Cloud Storage Systems
    - Use a storage device that is not connected directly to the user's computer
    - Network Storage - accessed through a local network
    - Cloud Storage – access via the Internet
      - Cloud applications provide online storage for their services (Google Docs, Flickr, Facebook, etc.)
      - Cloud storage services allow users to store documents online (Box, Google Drive, Microsoft SkyDrive, etc.)
        - Files can be accessed from anywhere via an Internet-enabled device
        - Files can be synched with your computer

79 STORAGE HARDWARE

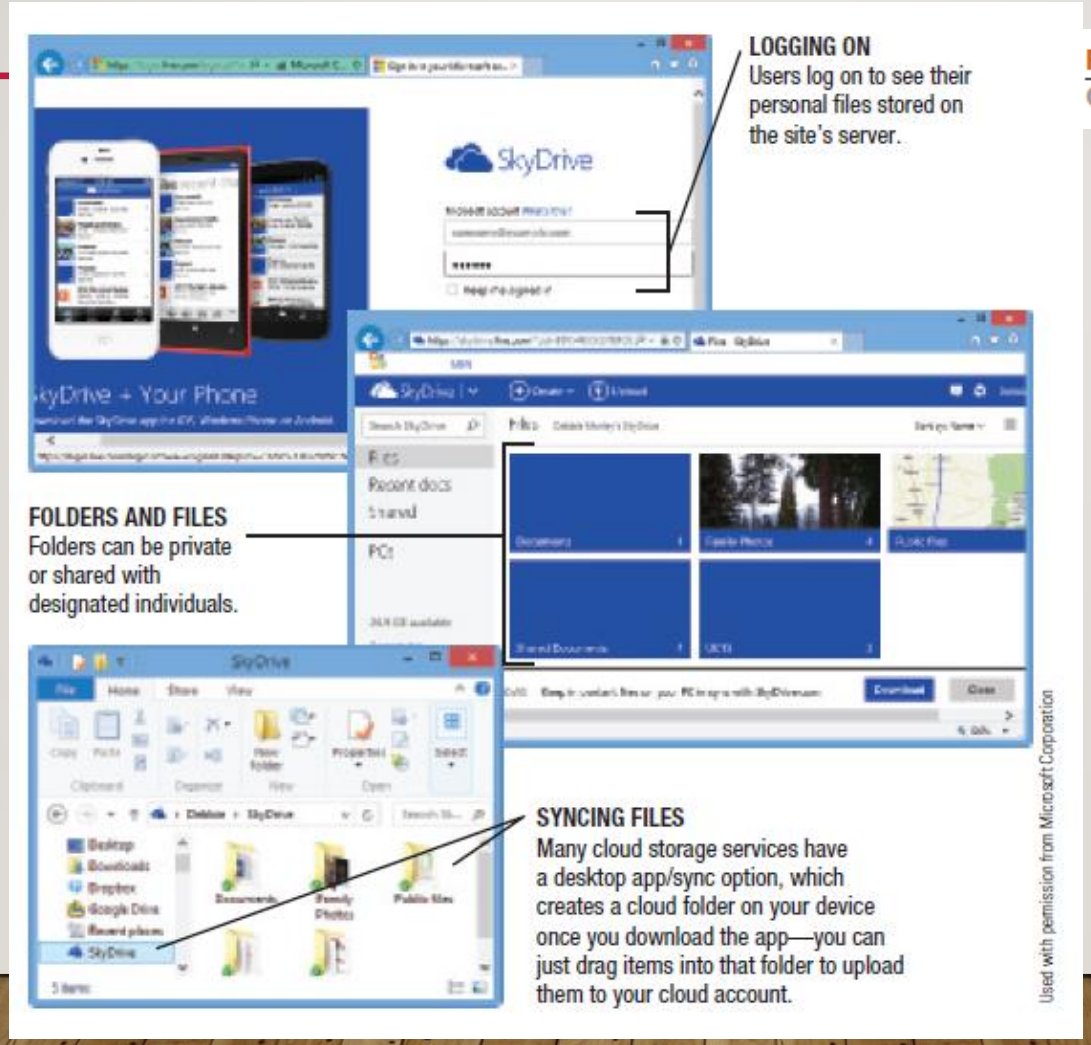


FIGURE 2-34
Cloud storage.

80 STORAGE HARDWARE

- Smart Cards
  - Credit card-sized piece of plastic that contains computer circuitry and components
  - Hold small amounts of data (64 KB or less)
  - Read by smart card reader
  - Often used for payment or identification purposes

FIGURE 2-35
Smart cards.



81 STORAGE HARDWARE

- Evaluating Your Storage Alternatives
  - Characteristics

Speed	Portability	Convenience
Price	Compatibility	Storage Capacity

  - Research which media and devices best fit your situation
    - Most computers will need a hard drive, a recordable or rewritable optical drive, a flash memory card reader, and at least one convenient free USB port

QUICK QUIZ

1. Of the following three options, the storage media that would likely hold the most data is a(n) _____.
    a. Internal hard drive
    b. USB flash memory drive
    c. Smart card
2. True or False: A DVD+RW disc can be written to, erased, and rewritten to as needed.
3. The circular rings on a magnetic disk on which data is stored are called _____.

Answers:

1) a; 2) True; 3) tracks

83 COMMUNICATIONS HARDWARE

- Communications Hardware
  - Enables the user to communicate with others over a network or the Internet
- The type of communications device used depends on:
  - The device being used (desktop computer, notebook, mobile phone, etc.)
  - The communication standard being used (such as Ethernet for wired networks, Wi-Fi for wireless networks, Bluetooth for short-range wireless connections, or a cellular standard for mobile phones)

84 COMMUNICATIONS HARDWARE

- Network Adapters and Modems
  - Network Adapter (network interface card (NIC))
    - Used to connect a computer to a network
    - The type of adapter used depends on the type of network and communications medium being used
  - Modem
    - Used to connect a computer to another computer or to a network (typically the Internet) over phone lines
    - Also used to refer to any device that connects a computer to a broadband Internet connection, i.e., cable modem

85 COMMUNICATIONS HARDWARE

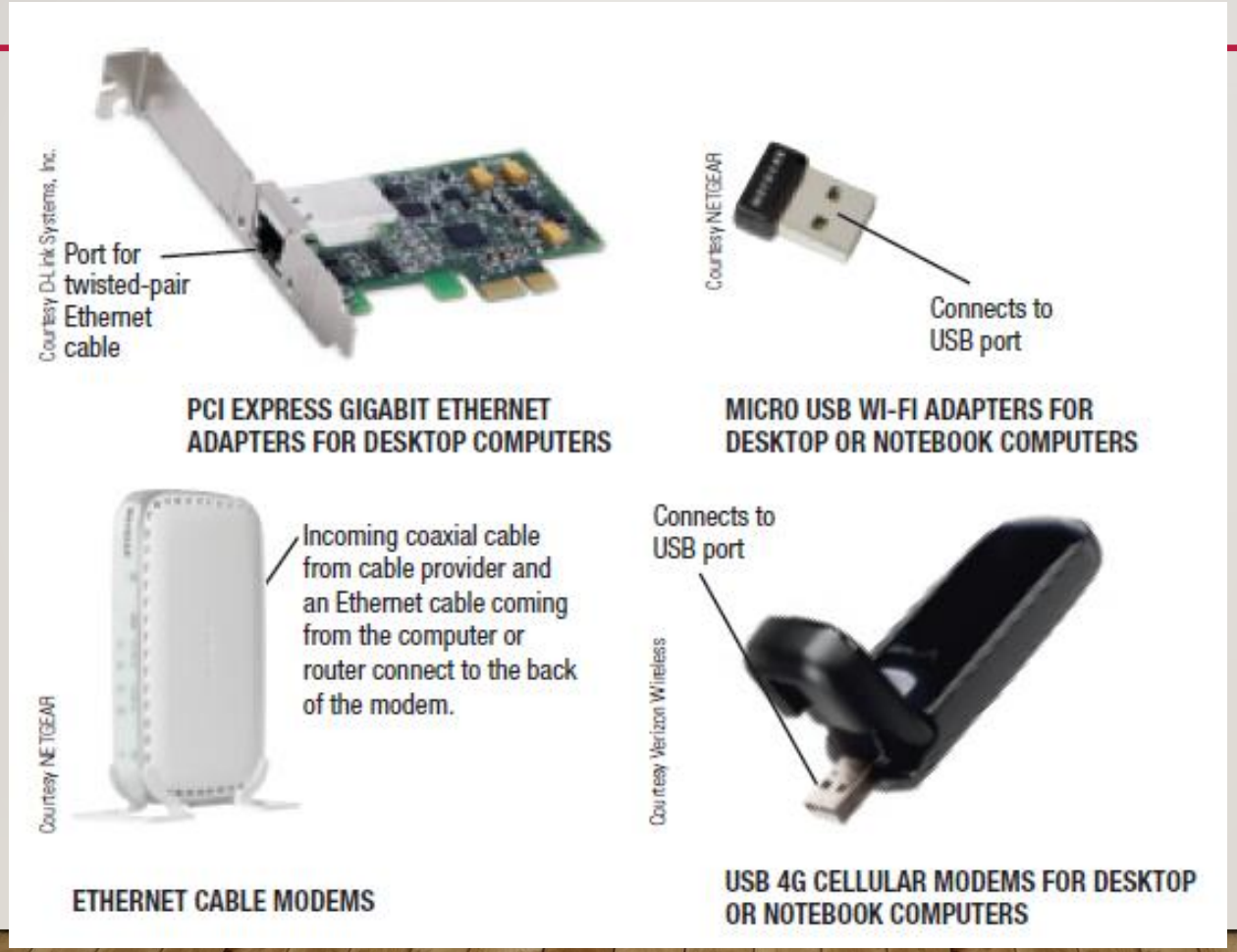


FIGURE 2-36
Network adapters and modems.

COMMUNICATIONS HARDWARE

- Each type of modem matches a particular type of Internet connection (conventional dial-up, cable, fixed wireless, DSL, etc.)
- Both network adapters and modems are available in internal and external versions and in a variety of formats (USB, Ethernet, etc.)

- Other Networking Hardware
  - Hub or switch
    - Central device used to connect devices in a wired network together

COMMUNICATIONS HARDWARE

- Wireless access point or wireless router
  - Central device for wireless networks
- Repeaters, range extenders, antennas
  - Used to increase the range of a network
- Cabling
  - Connects wired devices
  - Twisted-pair, coaxial cable, fiber-optic cable

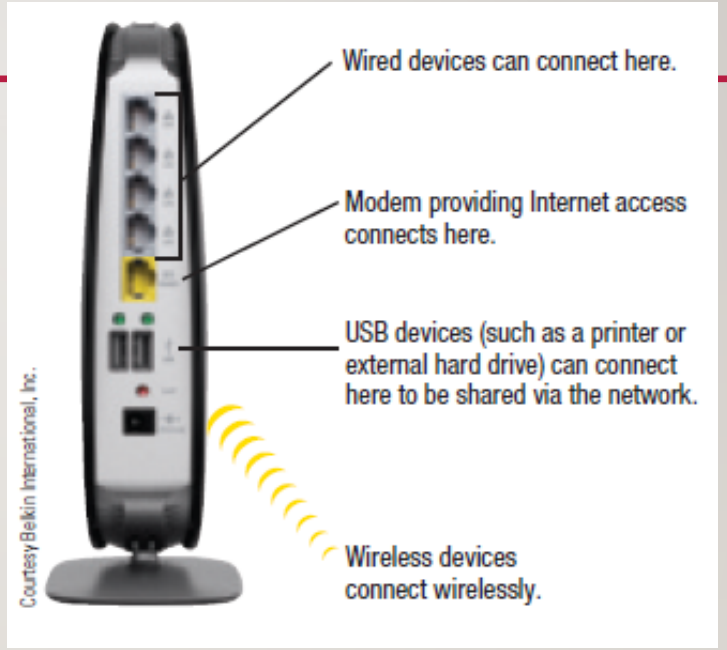


FIGURE 2-37
Wireless routers. Enable devices to connect to each other and to an Internet connection.

QUICK QUIZ

1. Which of the following is used to connect a computer to a network?
    a. Modem
    b. Network adapter
    c. Repeater
2. True or False: Both network adapters and modems are available in internal and external versions.
3. A wireless _____ is the central device for a wireless network.

Answers:

1) b; 2) True; 3) router; access point

89 SOFTWARE BASICS

- Software Ownership Rights
  - Specify the allowable use of a program
  - Software license
    - Explains conditions for use of the software, such as how many computers on which the program may be installed
  - Basic categories of software:
    - Commercial: Developed and sold for profit
    - Shareware: Distributed on the honor system
    - Freeware: Given away free of charge
    - Public domain: Not copyrighted
    - Open source software: Programs whose source code is available to the general public

SOFTWARE BASICS

- Desktop vs. Mobile Software
  - Desktop Software
    - Used by portable computers and desktop computers
  - Mobile Software (mobile apps)
    - Software specifically designed for a specific type of mobile phone or device
    - Typically has a more compact, efficient appearance than desktop software
    - Include features for easier data input, i.e., onscreen keyboard
    - Is usually downloaded

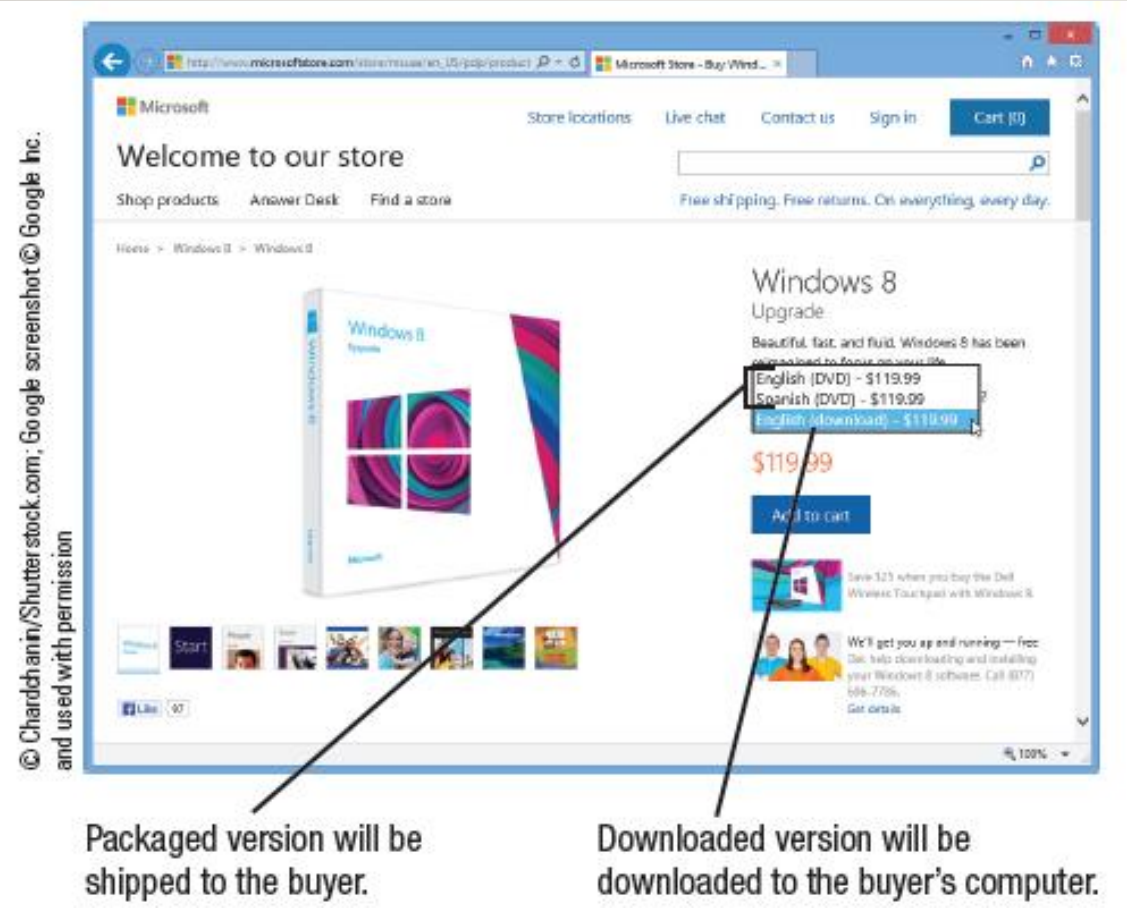


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FIGURE 2-38
Mobile software.

91 SOFTWARE BASICS

- Installed vs. Cloud Software
  - Installed Software
    - Installed on and run from the user's device
    - Can be purchased in physical form or downloaded
  - Cloud Software
    - Accessed on demand from any device with an Internet connection
    - Also referred to as Software as a Service (SaaS) or cloudware
    - May be slower than applications stored on the hard drive
    - Documents cannot be accessed if the server is down, or when you do not have Internet access



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FIGURE 2-39
Installed software.
Is often purchased via the Internet.

93 TREND BOX

Mobile App Builders

- Many tools are available to help develop mobile apps and deploy them on various platforms
- One example is appsbar
- After the app is created, appsbar tests it and then submits it to major app markets for publication



SOFTWARE BASICS

- Common Software Commands
  - Many concepts and commands are similar from app to app
  - Can be issued via menus, keyboard shortcuts, or command buttons on a toolbar or Ribbon
    - Menus
      - Text-based lists that provide access to commands that can be selected
    - Keyboard Shortcuts
      - Key combinations that correspond to specific commands
        - Usually combinations using the Ctrl or Alt keys, the function keys, and/or alphanumeric keys

SOFTWARE BASICS

- Toolbar
  - Contains buttons used to issue command
- Ribbon
  - Consists of tabs that contain groups of related commands
  - First introduced with the release of Microsoft Office 2007
    - Home tab contains the most frequently used commands
    - Contextual tabs are displayed as needed
    - File tab replaces the Office button

SOFTWARE BASICS

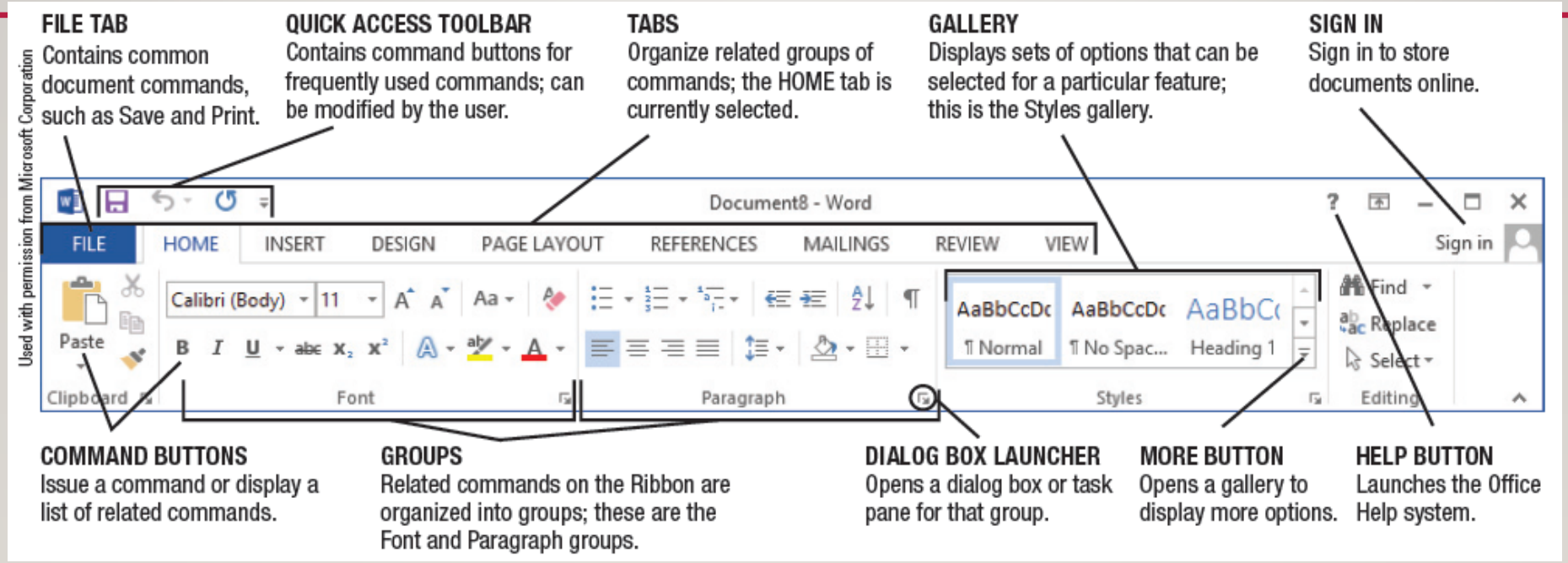


FIGURE 2-40
The Microsoft Office Ribbon.

97 SOFTWARE BASICS

- Editing a Document
  - Changing the content of a document, such as adding or deleting text
  - Insertion point marks the current location
  - Typing text adds it at that location
  - Delete and Backspace key are used to delete text
- Formatting a Document
  - Changing the appearance of a document
  - Changing the font face, size, color, style
  - Changing line spacing or page margins
  - Adding page numbering, borders, or shading

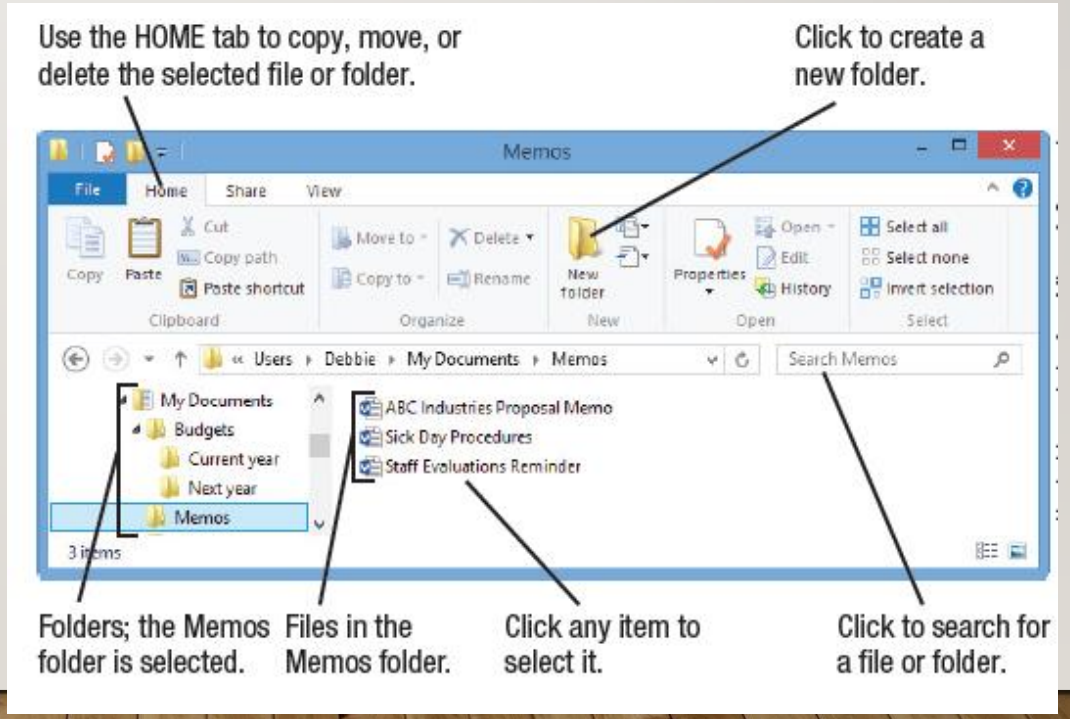
98 SOFTWARE BASICS

- Working with Files and Folders
  - File
    - Anything stored on a storage medium (program, letter, digital photo, song, etc.)
  - Filename
    - Name the user gives a file
  - Folders
    - Used to store related files to keep files organized
  - File management program
    - Allow you to perform file management tasks easily and efficiently (copying files, deleting files, etc.)

SOFTWARE BASICS

- Allow you to perform file management tasks easily and efficiently
  - View files
  - Copy and paste files
  - Delete files
  - Create folders to organize files
  - Search for files

FIGURE 2-41
File Explorer.



100 QUICK QUIZ

1. When you change the content of a document, such as adding or deleting text, you are _____.
    a. Editing the document
    b. Formatting the document
    c. Copying the document
2. True or False: The keyboard shortcut is the new tool available in recent versions of Microsoft Office to issue commands.
3. The file management program in Windows 8 is _____.

Answers:

1) a; 2) False; 3) File Explorer

101 SUMMARY

- Digital Data Representation
- Input Hardware
- Processing Hardware and Other Hardware Inside the System Unit
- Output Hardware
- Storage Hardware
- Communications Hardware
- Software Basics